CURRICULUM VITAE

ЛЕОНОВ ДЕНИС ВЛАДИМИРОВИЧ / DENIS LEONOV

PERSONAL DATA

Full Name	Леонов Денис Владимирович / Denis Vladimirovich Leonov
Date of Birth	3 November 1989
Citizenship	Russian Federation
Academic Degrees	• **Doctor of Technical Sciences (Dr.Sci.Tech.)** — 13.02.2026 • **PhD in Technical Sciences (Cand.Sci.Tech.)** — 19.04.2019
Current Positions	• Leading Scientist, Research and Practical Clinical Center for Diagnostics and Telemedicine Technologies, Moscow Health Care Department • Associate Professor (part-time), Moscow Power Engineering Institute (MPEI, National Research University)

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EDUCATION AND ACADEMIC DEGREES

Year	Degree	Institution	Thesis Title
2026	**Doctor of Technical Sciences (Dr.Sci.Tech)**	Research and Practical Clinical Center for Diagnostics and Telemedicine Technologies / Moscow Power Engineering Institute	**"Methods and means of improving the quality of ultrasound diagnostics in medicine"** (Defended: February 13, 2026)
2019	**PhD in Technical Sciences (Cand.Sci.Tech.)**	Moscow Power Engineering Institute (MPEI)	**"Ultrasound medical diagnostic system based on Doppler twinkling artifact"** (Defended: April 19, 2019)
2013	**Master of Science (MSc)**	Moscow Power Engineering Institute (MPEI)	**Biomedical Engineering**

CURRENT POSITIONS AND RESPONSIBILITIES

2024–present: Leading Scientist
Research and Practical Clinical Center for Diagnostics and Telemedicine Technologies, Moscow Health Care Department

- Leading research in ultrasound signal processing, beamforming, and quantitative ultrasound
- Supervision of 3 PhD candidates and 1 PhD student
- Development of novel methods for microcalcification detection using Doppler twinkling artifact
- Design and validation of anatomically realistic ultrasound phantoms

2020–present: Associate Professor (part-time)
Moscow Power Engineering Institute (MPEI, National Research University), Department of Radioengineering Fundamentals

- Teaching courses in digital signal processing and medical imaging
- **Academic supervisor for 7 undergraduate and master's students**
- Supervision of student research projects in ultrasound advanced signal processing and phantom design

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SUPERVISION OF PhD CANDIDATES AND STUDENTS

PhD Candidates and Students (Current)

Name	Status	Dissertation Topic	Expected Completion
Yulia Vladislavovna Bulgakova	PhD Student (2nd year)	"Advanced beamforming and aberration correction methods for improving ultrasound image quality in medical diagnostics"	2028
Anastasia Aleksandrovna Nasibullina	PhD Candidate (soiskatel)	"Development and validation of ultrasound phantoms with controlled acoustic properties for diagnostic applications"	2029

Name	Status	Dissertation Topic	Expected Completion
Veronika Vyacheslavovna Grebennikova	PhD Candidate (soiskatel)	"3D printing technologies for anatomically realistic ultrasound phantoms and their application in training and research"	2029
Olga Viktorovna Vlasova	PhD Candidate (soiskatel, final year)	"Digital signal processing methods for quantitative ultrasound analysis of breast tissue"	2027

Undergraduate and Master's Students

Level	Number	Institution	Role
Master's students	3	Moscow Power Engineering Institute (MPEI)	Research supervisor
Bachelor's students	4	Moscow Power Engineering Institute (MPEI)	Research supervisor

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KEY RESEARCH PROJECTS (Responsible Executor)

Period	Project Title	Registration Number	Role
2023–2025	R&D project "Development of design, manufacturing technology, and production of phantoms to increase the informativeness of ultrasound diagnostic studies"	№ ЕГИСУ: 123031500001-4	Responsible Executor
2023–2025	Research project "Scientific substantiation of the development	№ ЕГИСУ: 123092000013-3	Responsible Executor

Period	Project Title	Registration Number	Role
	and application of tissue-equivalent materials for the creation of test objects in the field of radiation diagnostics"		
2020–2022	Research project "Improvement of medical-technological and organizational aspects of ultrasound diagnostics"	№ ЕГИСУ: АААА-А20-120071090054-9	Responsible Executor
2017–2019	Research project "Improvement of methodology and medical-organizational aspects of ultrasound diagnostics"	№ ЕГИСУ: АААА-А19-119062890087-1	Responsible Executor

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PUBLICATIONS (Selected, 2018–2026)

Peer-Reviewed Journal Articles

1. **Leonov D.** , Zhou Z., Costa-Junior J.F.S., Yakovleva T., Vlasova O., Omelianskaya O., Vasilev Yu. Comparison of Three Aberration Correction Techniques in Transcranial Synthetic Aperture Ultrasound Imaging Using Experimental Wavefront Estimation. Biomedical Physics & Engineering Express. 12 (2026) 025059**.** https://doi.org/10.1088/2057-1976/ae5711 [IF: 1.6]
2. Grebennikova V., **Leonov D.** , Zhou Z., Costa-Júnior J.F.S., Shestakova D., Saikia M.J., Vetsheva N., Kulberg N., Pashinceva K., Omelianskaya O., Vasilev Yu. Design and validation of a technology for 3D printing training phantoms for ultrasound imaging. *Physical and Engineering Sciences in Medicine*. 2025. https://doi.org/10.1007/s13246-025-01670-7 [IF: 2.3]
3. Wang T., Zhu Q., Yu T., **Leonov D.** , Shi X., Zhou Z., Lv K., Xiao M., Li J. Multimodal deep learning-based classification of breast non-mass lesions using gray scale and color doppler ultrasound. *Diagnostics*. 2025;15(23):2967. https://doi.org/10.3390/diagnostics15232967 [IF: 3.6]
4. **Leonov D.** , Kulberg N., Yakovleva T. Aberration correction by polynomial approximation for synthetic aperture ultrasound imaging. *Medical Physics*. 2024;51(5):3292-3308. https://doi.org/10.1002/mp.17078 [IF: 3.8]

5. Yu T., Tsui P.H., **Leonov D.** , Wu S., Bin G., Zhou Z. LPC-SonoNet: A lightweight network based on SonoNet and light pyramid convolution for fetal ultrasound standard plane detection. *Sensors*. 2024;24(23):7510. https://doi.org/10.3390/s24237510 [IF: 3.9]
6. **Leonov D.** , Venidiktova D., Costa-Júnior J.F.S., Nasibullina A., Tarasova O., Pashinceva K., Vetsheva N., Bulgakova J., Kulberg N., Borsukov A., Saikia M.J. Development of an anatomical breast phantom from polyvinyl chloride plastisol with lesions of various shape, elasticity and echogenicity for teaching ultrasound examination. *International Journal of Computer Assisted Radiology and Surgery*. 2024;19:151-161. https://doi.org/10.1007/s11548-023-02911-4 [IF: 2.3]
7. **Leonov D.V.**., Kodenko, M.R., Leichenco, D.V. et al. Design and validation of a phantom for transcranial ultrasonography. Int J CARS 17, 1579–1588 (2022). https://doi.org/10.1007/s11548-022-02614-2
8. **Leonov D.V.** , Reshetnikov R.V., Kulberg N.S., Nasibullina A.A., Gromov A.I. Doppler twinkling artifact observations: an open-access database of raw ultrasonic signals. *Digital Diagnostics*. 2021;2(3):261-276.
9. **Leonov D.V.** , Kulberg N.S., Gromov A.I., Morozov S.P. Detection of microcalcifications using the ultrasound Doppler twinkling artifact. *Biomedical Engineering*. 2020;54:174-178.
10. **Leonov D.V.** , Kulberg N.S., Gromov A.I., Morozov S.P., Kim S.Y. Causes of ultrasound Doppler twinkling artifact. *Acoustical Physics*. 2018;64(1):105-114. https://doi.org/10.1134/S1063771018010128 [IF: 1.1]

CONFERENCE ORGANIZATION

Dr. Leonov has actively contributed to the organization of national and international scientific conferences:

Conference	Role	Location	Years
Moscow Oncological Forum	Organizing Committee Member	Gostiny Dvor, Moscow	2023, 2024, 2025
Online Diagnostics Conference	Organizing Committee Member	Moscow	2023-2026
Young Scientists Conference	Organizing Committee	Research and Practical Clinical	2022–2026

Conference	Role	Location	Years
	Member	Center for Diagnostics and Telemedicine Technologies, Moscow	
International Scientific and Technical Conference of Students and PhD Students "Radioelectronics, Electrical Engineering and Energy"	Session Organizer	Moscow Power Engineering Institute (MPEI)	2020–2026

INVITED TALKS AND PRESENTATIONS

- "Advances in ultrasound diagnostics: signal processing, phantom design, and clinical translation." *Seminar at Beijing University of Technology*, Beijing, China, **September 2024**.
- "Quantitative ultrasound and aberration correction: bridging engineering and clinical practice." *Indian Institute of Technology Hyderabad (IITH)*, Hyderabad, India, **November 2024**.
- "Ultrasound imaging and photoacoustics: synergies in phantom development." *Indian Institute of Science Education and Research Thiruvananthapuram (IISER-TVM)*, Thiruvananthapuram, India, **September 2025**.
- "Advances in ultrasound phantom development and aberration correction." *University of São Paulo (USP) Ribeirão Preto*, Ribeirão Preto, Brazil, **December 2025**.

TEACHING EXPERIENCE

Moscow Power Engineering Institute (MPEI)

Course	Level	Years	Role
Digital Signal Processing	Bachelor's	2020–present	Lecturer

Course	Level	Years	Role
Medical Imaging Systems	Master's (elective)	2021–present	Lecturer
Ultrasound Beamforming: Theory and Practice	Master's (elective)	2022–present	Lecturer
Research Seminar in Biomedical Engineering	Master's/PhD	2020–present	Co-coordinator

Student Supervision:

- 7 undergraduate and master's students (current)
- 4 PhD candidates/students (current)

INTERNATIONAL COLLABORATIONS AND NETWORK

Active Research Collaborations

Partner	Institution	Country	Focus Area	Joint Activities
Prof. Zhuhuang Zhou	Beijing University of Technology	China	Quantitative ultrasound, deep learning	Joint publications (2024-2025), student exchanges, annual workshops
Prof. Po-Hsiang Tsui	Chang Gung University	Taiwan	H-scan imaging, entropy imaging	Joint publications, collaborative research
Dr. José	Brazilian Air	Brazil	Ultrasound	Joint

Partner	Institution	Country	Focus Area	Joint Activities
Francisco Silva Costa-Júnior	Force Academy / University of São Paulo (USP) Ribeirão Preto		phantoms	publication (2024), phantom design collaboration, invited lecture (2025)
Prof. Mahesh Raveendranatha Panicker	Singapore Institute of Technology (SIT)	Singapore	Local speed of sound estimation, portable ultrasound scanner development	Joint conferences, research visits, student exchanges, collaborative projects on portable ultrasound systems
Prof. Nagarajan Ganapathy	Indian Institute of Technology Hyderabad (IITH)	India	AI for ultrasound image segmentation, deep learning architectures	Joint workshops, student exchanges, shared datasets for segmentation challenges, invited lecture (2024)
Prof. Suheshkumar	Indian Institute of	India	Aberration correction	Joint conferences,

Partner	Institution	Country	Focus Area	Joint Activities
Singh	Science Education and Research Thiruvananthapuram (IISER-TVM)		techniques, phantom development for ultrasound and photoacoustics	mutual research visits, collaborative phantom validation studies, invited lecture (2025)

Collaborative Activities

The collaboration with international partners extends beyond principal investigator interactions to include:

- **Student exchanges:** PhD students from both sides participate in short-term research visits (1-3 months)
- **Joint workshops:** Annual international workshops on ultrasound imaging (alternating between countries)
- **Shared datasets:** Collaborative development and sharing of ultrasound datasets for algorithm validation
- **Joint supervision:** Co-supervision of PhD students in selected projects

These collaborations have already resulted in joint publications [Yu et al., 2024; Grebennikova et al., 2025; Wang et al., 2025] and are expected to intensify during the proposed project.

PROFESSIONAL MEMBERSHIPS AND SERVICE

- Member, Russian Acoustical Society
- Reviewer for: Diagnostics, Sensors, BMC Medical Edusaction, BMC Research Notes, Acoustical Physics, Biomedical Engineering
- Technical Program Committee, Russian Conference on Biomedical Engineering (2024-2026)

INDUSTRIAL PARTNERSHIPS

Partner	Country	Focus	Collaboration Type
JSC "Zaslon"	Russia	Research ultrasound scanner development	Joint R&D, technology transfer, access to engineering expertise

LANGUAGES

Language	Proficiency
Russian	Native
English	Fluent (reading, writing, speaking)
German	Basic

Last updated: April 2026